PYQ Pseudo Code 2

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Q1) What will be the value of the following pseudocode?

```mathematica
Integer value, n, num
Set value = 1, n = 45
num = num >> 1
num = num + value
Print num
```

Options:

- A. 44
- B. 0
- C. 1
- D. 12

Q1 - Explanation / Dry Run

- Initially: `value = 1`, `n = 45`, `num` is declared but not initialized → default integer value = 0 (in typical pseudocode)
- `num = num >> 1` → `0 >> 1 = 0`
- `num = num + value` → `0 + 1 = 1`
- Print `num` → **1**

Answer: Option C

Q2) What will be the value of the following pseudocode?

vbnet

```
Integer x, y
for each x from 1 to 11
    x = x + 2
end for
Print x
```

Options:

- A. 11
- B. 10
- C. 12
- D. 13

Q2 – Explanation / Dry Run

- Loop variable `x` runs from 1 up to 11 inclusive.
- Inside the loop, `x = x + 2`. Also, the loop update itself will increment `x` by 1 at the end of iteration (as per the “for each” semantics)
- Effectively, in each full iteration `x` increases by 3 (one from loop increment + two from inside)
- Starting at 1 → next becomes 4 → then 7 → then 10 → then 13 → loop stops when `x` > 11
- After loop ends, `x = 13`
- Print `x` → **13**

Answer: Option D

Q3) What will be the value of the following pseudocode?

mathematica

```
Integer j, m
Set m = 1, j = 1
Integer a[3] = {0, 1, 0}
a[0] = a[0] + a[1]
a[1] = a[1] + a[2]
a[2] = a[2] + a[0]
if (a[0])
    a[j] = 5
End if
m = m + a[j]
Print m
```

Options:

- A. 3
- B. 2
- C. 6
- D. 4

Q3 - Explanation / Dry Run

- Start: `m = 1, j = 1, a = {0,1,0}`
- `a[0] = a[0] + a[1]` → `0 + 1 = 1`
→ now `a = {1,1,0}`
- `a[1] = a[1] + a[2]` → `1 + 0 = 1`
→ `a = {1,1,0}` (no change really)
- `a[2] = a[2] + a[0]` → `0 + 1 = 1`
→ `a = {1,1,1}`
- `if (a[0])` → since `a[0] = 1` (true), execute `a[j] = 5` → `a[1] = 5` → `a = {1,5,1}`
- `m = m + a[j]` → `m = 1 + a[1] = 1 + 5 = 6`
- Print `m` → **6**

Answer: Option C

Q4) What will be the value of the following pseudocode for k = 150?

kotlin

```
fun(integer k)
    if (k > 155)
        return
    end if
    print k
    fun(k + 2)
    print k
End fun
Print output
```

Options:

- A. 150 152 154
- B. 150 152 154 154 152 150
- C. 150
- D. None

Q4 - Explanation / Dry Run

- Call `fun(150)`
  - Check `k > 155`? → false
  - Print `150`
  - Call `fun(152)`
    - Print `152`
    - Call `fun(154)`
      - Print `154`
      - Call `fun(156)`
        - Here `k = 156` → `k > 155` is true → return immediately
      - After recursion returns, print `154`
    - After that, print `152`
  - After that, print `150`
- Whole printed sequence: **150 152 154 154 152 150**

Answer: Option B

Q5) What will be the output of the following pseudocode?

pgsql

```
Integer a[5], b[5], c[5], k, l
Set a[5] = {5, 9, 7, 3, 1}
Set b[5] = {2, 4, 6, 8, 10}
for each k from 0 to 4
    c[k] = a[k] - b[k]
end for
for each l from 0 to 4
    Print c[l]
end for
```

Options:

- A. 7 13 13 11 11
- B. 3 5 1 -5 -9
- C. -3 -5 -1 5 9
- D. None

Q5 - Explanation / Dry Run

- Initialize arrays `a = {5, 9, 7, 3, 1}` and `b = {2, 4, 6, 8, 10}`
- Compute `c[k] = a[k] - b[k]` for each `k`:
  - `c[0] = 5 - 2 = 3`
  - `c[1] = 9 - 4 = 5`
  - `c[2] = 7 - 6 = 1`
  - `c[3] = 3 - 8 = -5`
  - `c[4] = 1 - 10 = -9`
- Print each element of `c`: `3 5 1 -5 -9`

Answer: Option B

Q6) How many times “A” will be printed in the following pseudocode?

vbnet

```vbnet
Integer a, b, c
for a = 0 to 4
    for b = 0 to 2
        if a > b
            Print "A"
        end if
    end for
end for
```

Options:

- A. 12
- B. 10
- C. 9
- D. 8

Q6 - Explanation / Dry Run

- Outer loop (`a` from 0 to 4) and inner loop (`b` from 0 to 2)
- The condition `if a > b` is true when:
  - For `a = 1`: `b = 0` → 1times
  - For `a = 2`: `b = 0, 1` → 2 times
  - For `a = 3`: `b = 0, 1, 2` → 3 times
  - For `a = 4`: `b = 0, 1, 2` → 3 times
- Total: 1 + 2 + 3 + 3 = **9**

Answer: Option C

Q7) What will be the output of the following pseudocode?

sql

```
Integer x, y
Set x = 3, y = 5
if x < y
    Print "X is less than Y"
else
    Print "X is greater than Y"
end if
```

Options:

- A. X is less than Y
- B. X is greater than Y
- C. X is equal to Y
- D. None

Q7 - Explanation / Dry Run

- Initialize `x = 3, y = 5`
- Condition `if x < y` → true
- Print `"X is less than Y"`

Answer: Option A

Q8) What will be the output of the following pseudocode?

```sql
Integer a, b
Set a = 10, b = 20
if a == b
    Print "Equal"
else
    Print "Not Equal"
end if
```

Options:

- A. Equal
- B. Not Equal
- C. Error
- D. None

Q8 - Explanation / Dry Run

- Initialize `a = 10, b = 20`
- Condition `if a == b` → false
- Print `"Not Equal"`

Answer: Option B

Q9) What will be the output of the following pseudocode?

vbnet

```
Integer i
for i = 1 to 5
    if i == 3
        continue
    end if
    Print i
end for
```

Options:

- A. 1 2 3 4 5
- B. 1 2 4 5
- C. 1 3 5
- D. None

Q9 - Explanation / Dry Run

- Loop `i` from 1 to 5
- When `i == 3`, `continue` skips the print statement
- Output: `1 2 4 5`

Answer: Option B